

The Utility Problem

There are multiple reasons why load forecasting is an important activity for any electrical utility.

Resource planning & Energy Management - electricity cannot be easily stored on a large scale, and the energy produced must be consumed immediately. Accurate load forecasting allows utilities to plan better for future energy generation and distribution needs. This can help to ensure that there is enough power available to meet demand and avoid blackouts or brownouts. In addition, by forecasting future load, utilities can make more informed decisions about when to generate and purchase power and when to use energy storage systems. This can help to reduce costs and optimize energy usage.

Asset management - forecasting is important in terms of knowing whether the power system will be able to accommodate new demand (or whether there will be overloading in some areas). It is not reasonable to reinforce the grid only when the maximum capacity is exhausted because of the long investment horizons involved in reinforcing the grid (construction of a transformer station, construction of a new line including all permitting processes, etc.).

Price forecasting - accurate load forecasting can also help utilities to predict changes in energy prices. This can help them to make better decisions about buying and selling energy on the wholesale market.

Maintenance scheduling - load forecasting can help utilities to schedule maintenance of power generating and distribution infrastructure. This can prevent breakdowns and improve the reliability of the power supply. Additionally, this helps utilities conduct planned outages during times with the lowest impact.

Compliance & Regulatory relations - utilities need to comply with various regulations and standards, and accurate load forecasting helps them to ensure compliance with these regulations.

Customer service - by forecasting load patterns, utilities can also predict high-demand periods and proactively communicate with customers about potential service disruptions or design customer motivation opportunities for shifting the load.

Overall, accurate load short-term or long-term forecasting is critical for utilities to operate efficiently and effectively, maintain a reliable power supply, and reduce costs. For these reasons, it is important to have tools and processes which allow for such accurate forecasting.

The Data Solutions and Methods

There are several methods for electrical load forecasting, including:

- Time-series analysis - this method uses historical data to make predictions about future electrical loads. It can be further divided into methods such as moving averages, exponential smoothing, and ARIMA models.
- Artificial neural networks - this method uses a type of machine learning algorithm to model the relationship between input variables (such as temperature and day of the week) and output variables (electrical load).
- Support vector regression - this is another machine learning method that uses a set of input-output pairs to train a model that can predict future electrical loads.
- Econometric models - this method uses economic and demographic data to predict future electrical loads.
- Hybrid models: This method combines multiple forecasting methods (like the ones mentioned above) to improve the accuracy of predictions.

The choice of method depends on the specific requirements and available data. With the Awesense Energy Transition Platform, it is possible to put into practice any of the mentioned methods since Awesense Platform offers a comprehensive and powerful set of data access methods for demanding analyst needs.

The following description will focus on a simple time-series analysis using the moving averages. The main advantage of moving averages is that it is simple to implement and understand, but they can be less accurate than other time-series analysis methods when the data contains a lot of fluctuations or if the window size is not appropriate. The described method will employ a simple moving average¹ (SMA) on the resolution of a feeder. The SMA method calculates the average of the past n observations. The number of past observations used in the calculation is called the window size. For example, if the window size is 3, the SMA would be the average of the past three observations.

To obtain the load data for a feeder, analysts using the Awesense Platform have two options:

1. Getting the time-series data directly from the top-of-the-feeder element (usually breaker) `ts_data_source_select()`.
2. If the time-series measurements are unavailable, analysts can infer the data by aggregating the downstream load. This can be accomplished by iterating through top-of-the-feeder elements and fetching the elements “with loads” (usually meter elements). This can be achieved by `grid_get_downstream()` or `grid_get_connected()` functions. After the meters are fetched, `ts_data_source_select()` will yield a consumption time-series, which can be aggregated and the feeder load can be calculated.

¹ Other moving averages are possible as well, specifically Weighted Moving Average (WMA) or Exponentially Weighted Moving Average (EWMA). Both put different emphases on more recent data and how the most recent data contribute to the overall result. The choice of moving average technique and window size depends on the specific requirements and characteristics of the data.

Load Forecasting

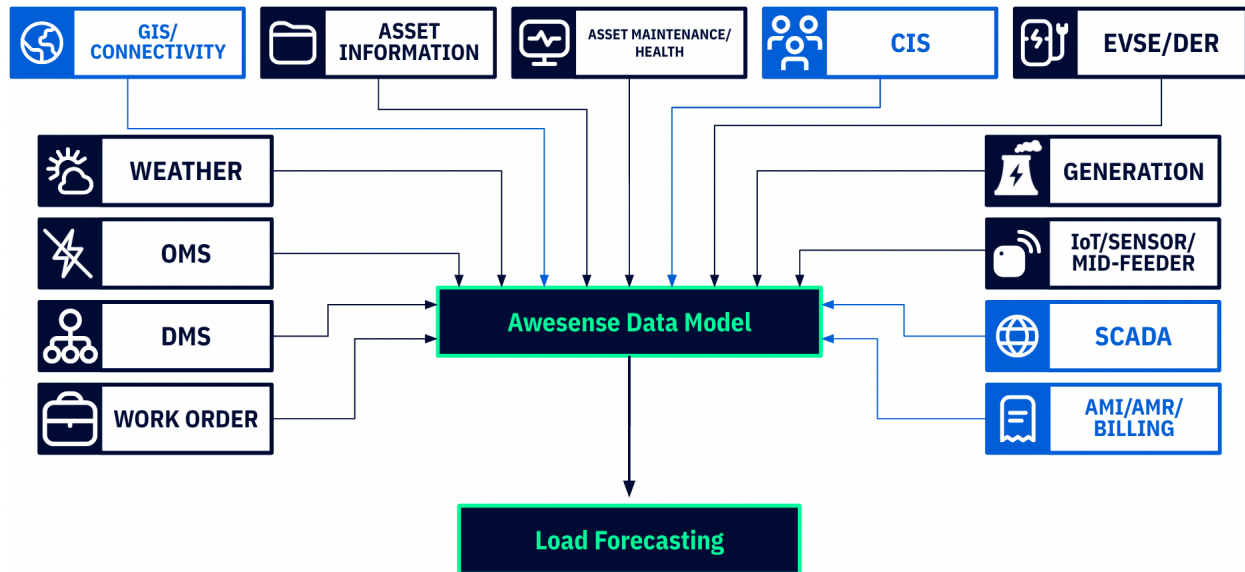
[UC43-01] - Awesense Energy Transition Platform Use Case



With historical feeder level load time-series, analysts can now employ the SMA method to calculate the load forecasting. As we mentioned earlier, any of the load forecasting methods can be used. A viable combination of SMA would be with economic models, which might bring some variables by geolocation into the calculation.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.